

Brick and mortar

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In November last year, we tested motherboards with integrated graphics (IG) solutions. These were strictly entry-level to mid-range platforms. As you're probably aware, high-end systems mostly mean motherboards without any graphics solution. Yes, high-end PCs demand a graphics card, and therefore, an integrated video solution isn't required. Moreover, most manufacturers reserve the top-end features and top quality binned parts for their top-of-the-line solutions. These are more-often-than-not discrete graphics solutions.

When you're simply buying a PC, any motherboard will most likely do, as long as it supports the processor of your choice – at least four serial ATA ports and an equal number of memory slots and USB ports. The motherboards we've tested are catered more towards the discerning user. Most

High-end rigs, multi-GPU systems and world record beating benchmarks need the best boards. We've tested them...

of these boards support extra connectivity options like E-SATA, extra USB ports and serial ATA ports. Some of these even come with a lot of snazzy add-ons, like an LCD screen that displays POST (Power On Self Test) messages. Perhaps more importantly, most of these motherboards support specific multi-GPU solutions like NVIDIA's SLI or ATI's Crossfire. If you're looking at building a fast gaming rig, or even a powerful multi-purpose PC, or a rig for cracking world benchmark records, then you'll probably end up shopping for one of these worthies.

We've tested 26 of the top-end boards available today. To nobody's surprise, 24 of these boards were Intel LGA 775 platforms. AMD users have a measly two boards to choose from. This is because most vendors are flocking to make high-end solutions supporting Core 2 Duo and Core 2 Quad processors, thanks to Intel's

domination in the high-end performance market. AMD chose to play the entry-level market, which is why our integrated-graphics motherboard comparison in November had more AMD-based solutions. If you're buying a high-end PC today, market trends suggest you want an Intel processor, and motherboard manufacturers seem to agree. We also noticed that Nehalem boards were rather scarce – it seems Intel's DX 58 SO is the only X58 motherboard that is readily available. Other vendors have demo pieces, but nothing in-market. We did get an MSI X58 Eclipse and an ASUS P6T Deluxe, but the former was recalled before we could test it. The latter turned out to be physically damaged – some CPU pins were missing. Therefore, Intel's X58 offering remains the only Nehalem-based board we've tested and we await more boards from other vendors based on the new and exciting Nehalem processor.

Motherboards up to Rs 10,000

Feature rich and fancy-free

Normally, non-integrated motherboards aren't cheap. However, there're always options for every user and in every price bracket. Sadly, the recent plummeting Rupee value against the Dollar means that most Indian distributors have pulled up their rates. Over the past two months, quite a few sub-Rs. 10000 boards have crossed into the higher price category. Also cheaper IG boards are plentiful, but non-integrated boards are one category where discerning buyers are willing to shell out big. As a result, only four boards made it into this cat-

egory and one was an AMD-based offering.

Features

The MSI K9A2 Neo2 was the AMD-based solution in this category, and is based on AMD's RX780 chipset. This board is cleanly laid out and the area around the CPU is also very open. Two of the others were P43-based solutions, which are lower variants of the P45 chipset. MSI's P43 Neo and Gigabyte's EP43-DS3L both featured the ICH 10 Southbridge. ECS was the only vendor to provide a P45 offering – the P45T-AD3. Quite surprisingly this board supported DDR3 memory; this is kind of contradictory to the product pricing as DDR3 memory is costly enough to negate any price benefits the

board brings. This board also has an option in the BIOS for enabling Intel's Robson, which by definition uses a small amount of cache on the board itself to speed up HDD operations. Unfortunately this board doesn't have any such cache and may rely on a flash drive for this purpose. We haven't come across this setting in the BIOS' of other boards before so



MSI P43 Neo

How We Tested

We categorised our motherboards into three price categories

1. Boards up to Rs. 10,000
2. Boards above Rs. 10,000 and up to Rs. 18,500
3. Boards above Rs. 18,500

Our Test system for the Intel platform consisted of a Core 2 Extreme QX 9650 (3.0 GHz) processor and 4 GB of RAM. For DDR2 boards we used Corsair's Dominators rated at 1066 MHz at timings of 4-4-4-12 at a speed of 800 MHz. For the DDR3 motherboards we used 4 GB of OCZ XTC Platinum clocked at 1333 MHz at timings of 9-9-9-20. Our graphics card of choice was ASUS' EN GTX 280 TOP; an NVIDIA GeForce GTX 280-based card whose core is factory overclocked to 670 MHz. Western Digital Raptor hard drives were our storage component; these have the spindle running at 10,000 rpm and are suitable for such high-end boards.

For AMD systems we used the Phenom 9950 processor clocked at 2.6 GHz. We used the same memory, graphics card and HDD for the AMD motherboards as well; this is done to minimise deviations in scores due to different components being used. By varying only the motherboard and the processor we are able to keep other variances down to a minimum. We used the latest motherboard BIOS' and the latest drivers that we downloaded from the respective sites. Our OS of choice was the tried and tested Windows XP Prof with service pack 2 installed. This is probably the last time we're testing boards on this ultra stable and reliable platform.

Features

We looked very carefully at overclocking options that these boards provide. With overclocking becoming very idiot-proof, even idiots (pardon the pun!) can play around with BIOS settings safely. Enthusiasts will also find a lot of valuable information as far as CPU and memory voltage parameters go. We rated these boards on the basis of both overclocking and overvolting options and how easy and transparent the settings in the menus are to this effect. We also looked at the features and connectivity options that these boards provide. The presence of digital audio ports,

E-SATA, dual LAN ports, solid state capacitors and the number of expansion slots available for PCI and PCIe cards was rated.

Performance testing

PC Mark 2005: This benchmark runs 48 discrete tests to test each performance related component subsystem – the processor, memory, video and storage. Finally it throws out a final score – which is a weighted average of the individual results.

SiSoft Sandra 2008 Lite: Sandra 2007 had specific benchmarks to test all the subsystems of a PC. It tests a CPU's floating point crunching ability under both arithmetic and multimedia applications. The application supports multiple cores and scales well in this regard. It also tests the read and write performance of the storage subsystem, and the memory bandwidth.

3D Mark 2006: Our graphics benchmark of choice.

CineBench R10: Tests the CPUs rendering ability; scales well with multiple cores.

WinRAR 3.8: Tests compression/decompression at the CPU level

Video encoding (DivX 6.8): is a CPU intensive test and using it while testing a motherboard tells us exactly how optimised the board in question is. The latest version 6.8 also has support for the new SSE4 instruction set.

File transfer test: This tells us a lot about Southbridge performance and data path optimisations on each motherboard, and how fast data is moving around on your hard disk.

Game benchmarks: Doom 3, S.T.A.L.K.E.R Shadow Of Chernobyl and Prey were our games of choice. We run these games at maximum settings for a good GPU benchmark and at minimum settings to get a sound CPU benchmark.

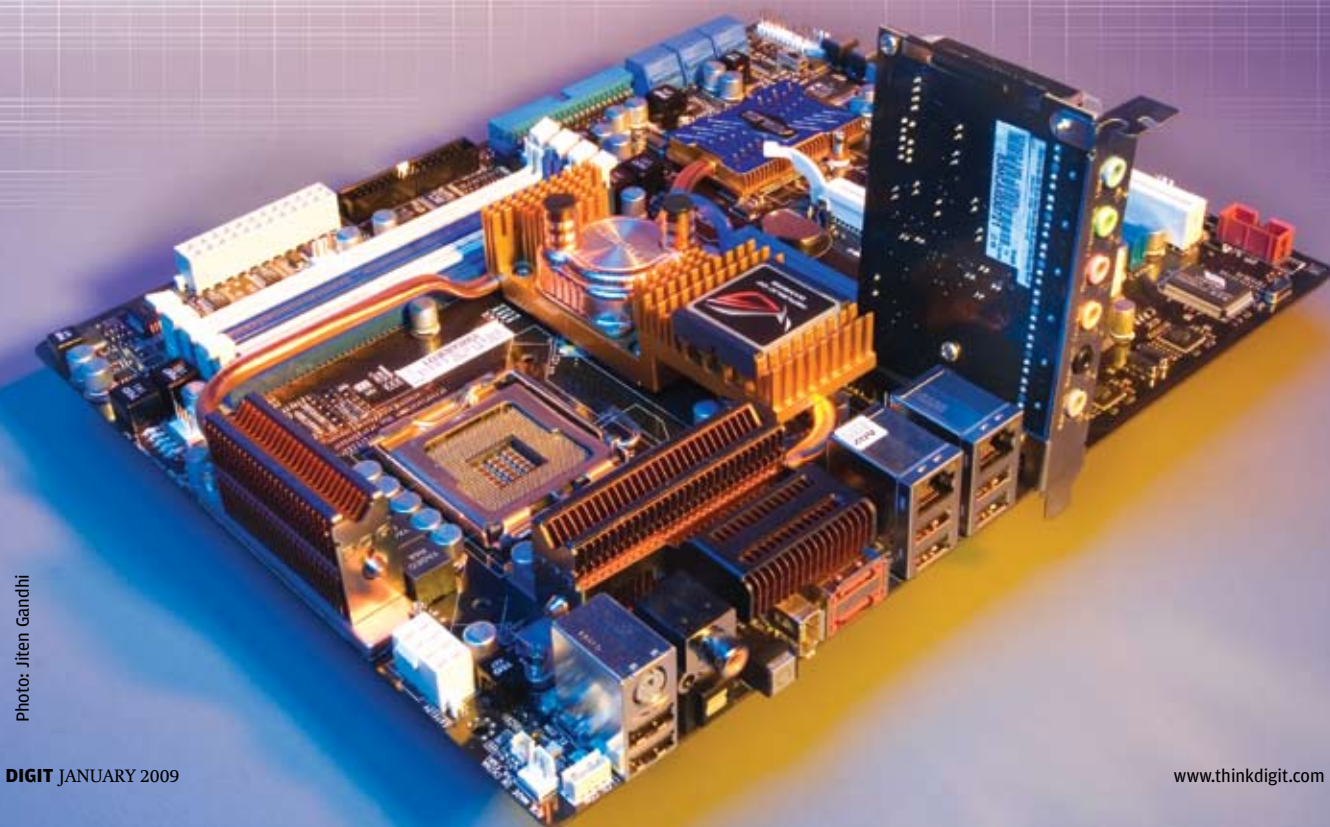


Photo: Jiten Gandhi